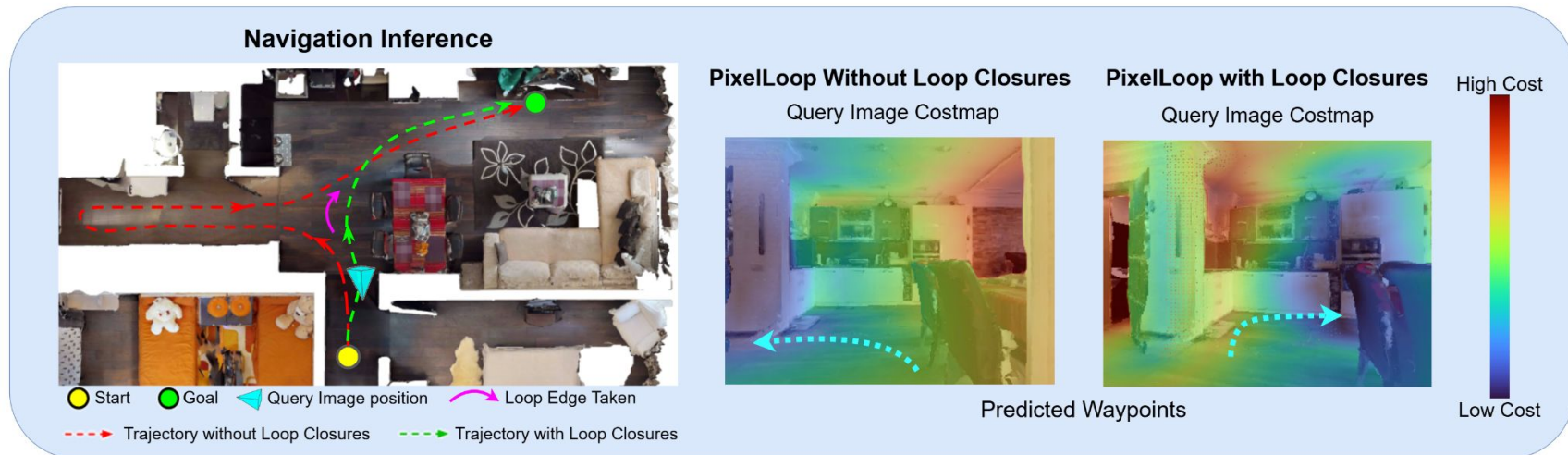


PixelLoop: Shortcut Topological Navigation with Pixel-Level Loops



Authors: Sarthak Chittawar, Vansh Garg, Aditya Vadali, Krish Pandya, Rohit Jayanti, Sourav Garg, K Madhava Krishna



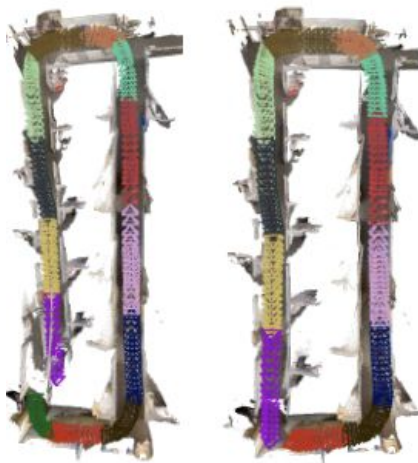
[pixelloop-nav.github.io](https://github.com/pixelloop-nav)

Problem Statement

- Current topological maps connect only **temporally adjacent observations**.
- Physically nearby locations often remain disconnected, forcing long detours.
- Image-Level or Object-Level loop closures provide coarse connectivity, but **lack the geometric precision** needed for reliable arbitrary start-to-goal navigation.
- Sequential topology therefore limits efficient **arbitrary start-to-goal navigation**, motivating **pixel-level loop closures** that recover shortcut connectivity **without requiring global metric reconstruction**.

Why Are Existing Loop Closures Insufficient?

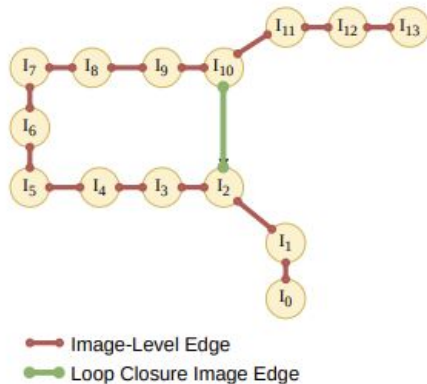
3D Map Based Methods



Without Loop Closure

With Loop Closure

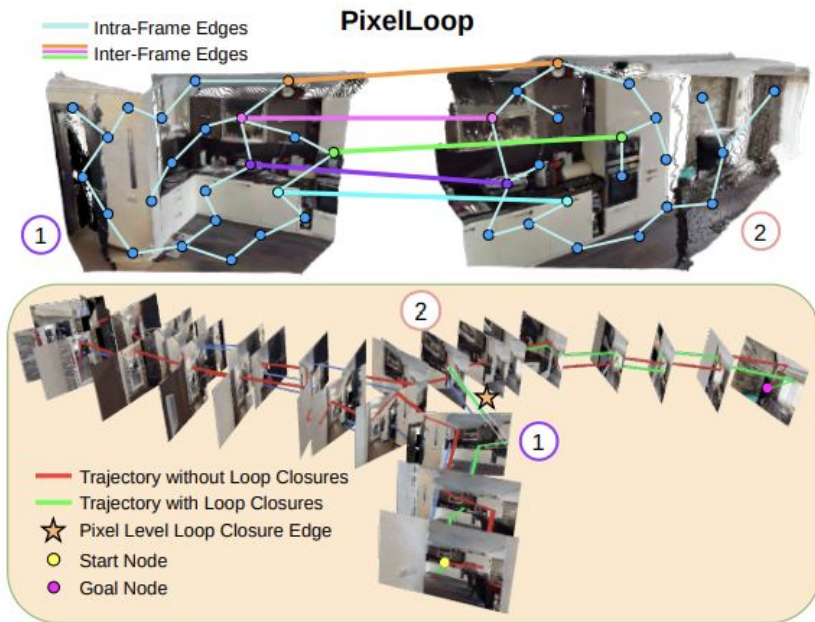
Image Topological Methods



Metric SLAM: Loop Closures used to correct drift

Image-/Object-Relative topologies: Sparse loop edges

Paper Objectives



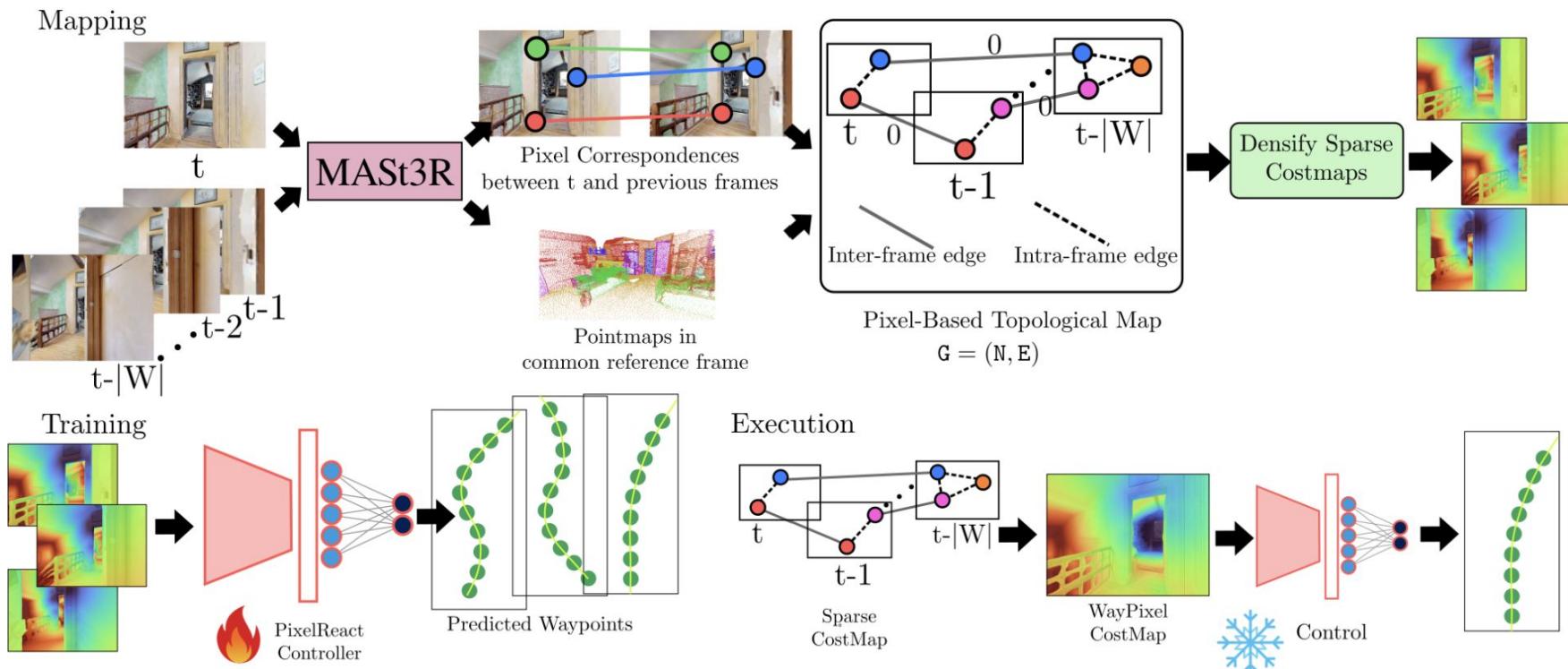
Introduce **dense Pixel-Level Loop Closures** in topological maps to improve goal-directed planning.

Enable **arbitrary start-to-goal navigation** in a Pixel-Relative topological graph with added loop edges, building on MAST3R-Nav.

Benchmark planning & navigation metrics against **Image-Level** & **Object-Level** loop closures.

Background Work

Pixel-level Topological Mapping (MASt3R-Nav)



Pixel-level Topological Mapping (MASt3R-Nav)

Builds a pixel-level topological graph from exploration images connected by temporally adjacent observations.

Uses dense pixel correspondences between the current observation and the localized image.

Generates a dense local **“WayPixel” costmap** for predicting controller waypoints during navigation.

Limitations:

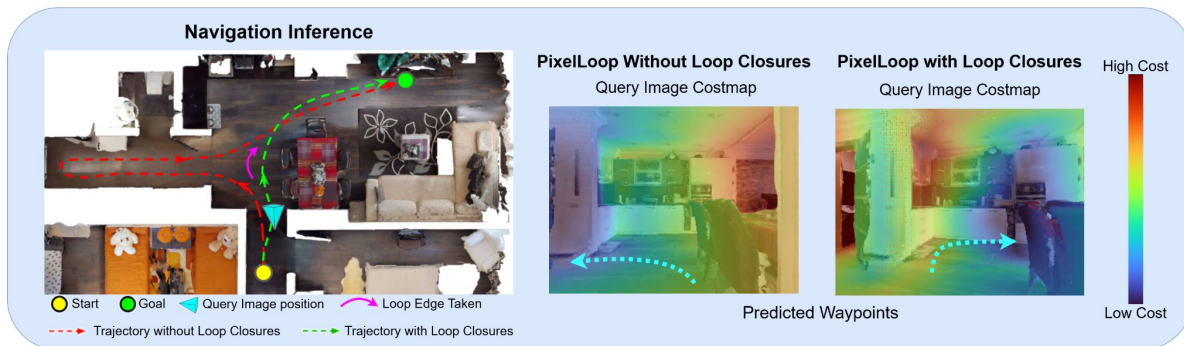
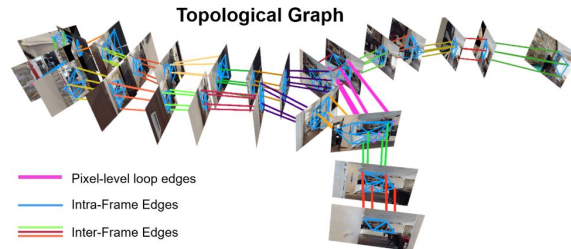
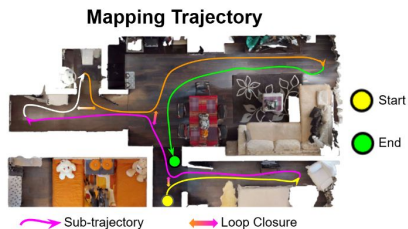
- Sequential topology limits efficient arbitrary start-to-goal navigation.
- Physically nearby locations remain disconnected without pixel-level loop closures.

Methodology

PixelLoop: Overview

PixelLoop implements & introduces:

- Sequence-based Visual Place Recognition
- Geometric loop verification
- Pixel-level loop closures to enable arbitrary start-to-goal navigation



Result:

- Shorter navigation paths
- Better localisation
- Higher navigation success rates

Why Sequence-Based VPR?

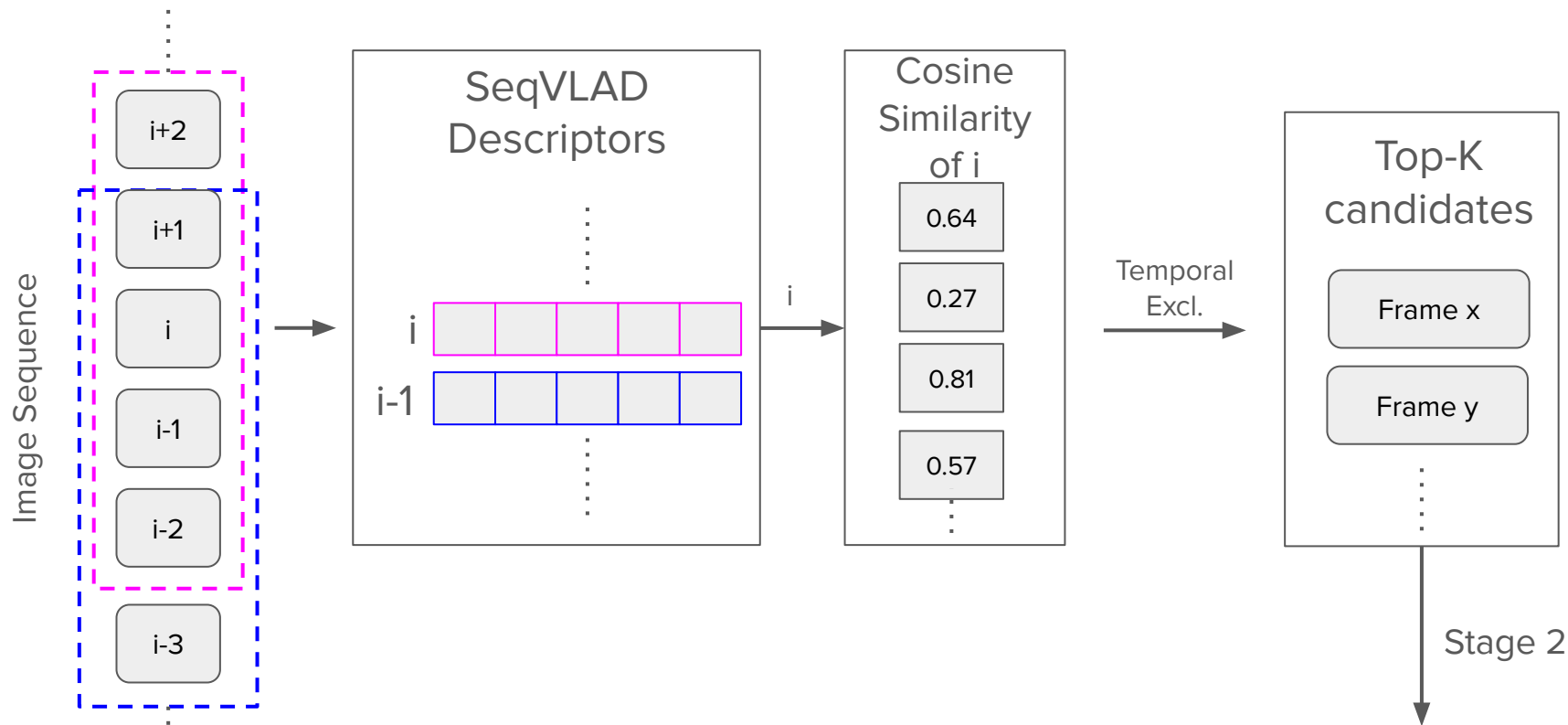
Challenge: Perceptual Aliasing in indoor environments (Eg: Identical shelves, curtains, limited unique visual landmarks)

In sequence based retrieval, we match a sequence of 5 images to the closest sequence of 5 images throughout the run.

This is done via SeqVLAD descriptors aggregated over a sliding window of 5 frames in the mapping run.

Key Benefits: Sequence descriptors significantly reduce false matches in repetitive indoor environments like in warehouses. Temporal context captures motion history and room context before and after a location.

Stage 1: Sequence Retrieval using SeqVLAD

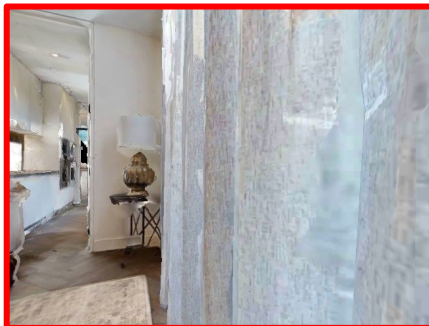


Sequence-Based VPR: Examples

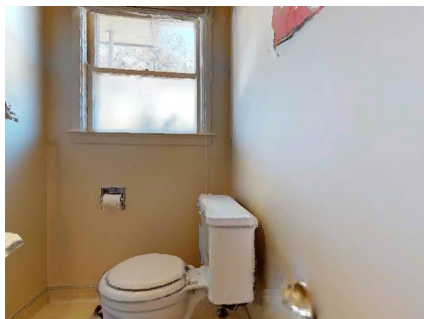
Query Img



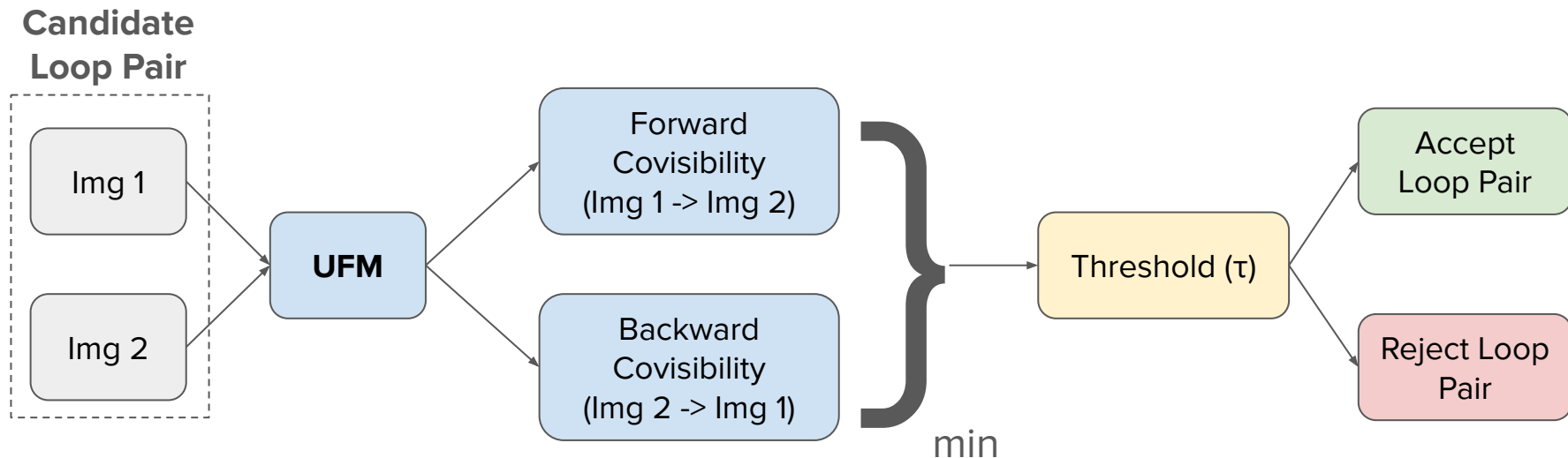
Incorrect Localised Img
(without sequence matching)



Correct Localised Img
(with sequence matching)



Stage 2: Geometric Verification using UniFlowMatch



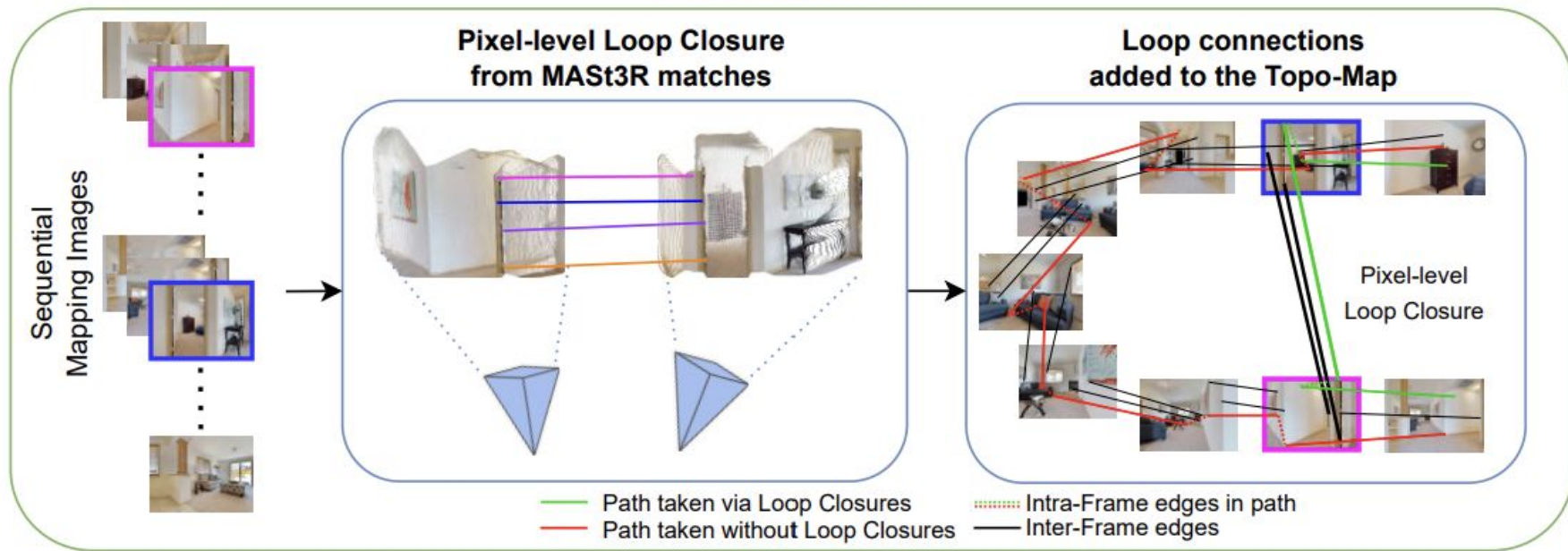
PixelLoop: Topological Graph

PixelLoop: Shortcut Topological
Navigation with Pixel-Level Loops

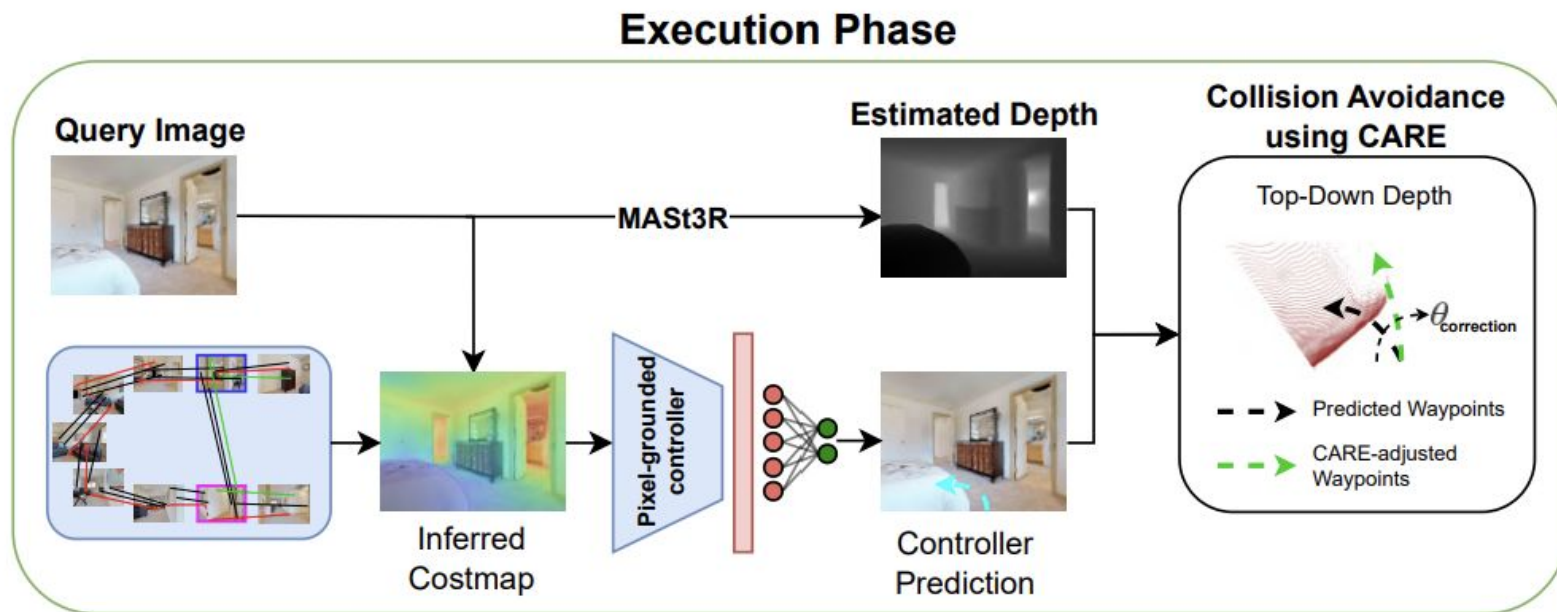
Project Page: <https://pixelloop-nav.github.io/>

PixelLoop Navigation Pipeline: Mapping Phase

Mapping Phase



PixelLoop Navigation Pipeline: Execution Phase



Experiments & Results

Experimental Setup

Simulation (Single-Traversal) Evaluation

- Habitat-Matterport 3D (HM3D)
- 6 indoor environments with 3 goals in each
- Average trajectory length: ~100 m
- 73 start-to-goal navigation tasks

Real-World Evaluation

- Pioneer P3DX mobile robot
- Intel RealSense RGB-D camera
- ROS Noetic middleware

Costmap Evaluation

- **Geodesic / Navigation Mesh-Based Costmaps:**

- Using Habitat-Sim, we compute geodesic distances from each pixel to the goal node via the navigation mesh.
- This provides an approximation of true shortest-path distances in the environment.

- **Ground Truth Pixel Correspondences:**

- We construct ideal costmaps using perfect pixel correspondences derived from globally registered 3D geometry.
- To maintain fair comparisons, we subsample these matches using farthest-point sampling to maintain the exact same density as the MAST3R model's predictions.
- This isolates planning errors from correspondence noise.

Costmap Evaluation

- We measure the similarity of PixelLoop's costmaps by comparing the MAE between ours and GT references.
- Specifically, the **MAE between binary masks (ours and GT)** formed by selecting pixels corresponding to the minimum $k\%$ of predicted cost values, for $k \in 5, 15, 30, 50$.
- Two forms of GT costmaps used:
 - **Geodesic / Navigation Mesh-Based Costmaps:** This provides an approximation of true shortest-path distances in the environment.
 - **Ground Truth Pixel Correspondences:** This isolates planning errors from correspondence noise.

Costmap Evaluation

Bold: Highest Overall

Underlined: Highest non-GT score

((a)) Geodesic / Navigation Mesh-Based Costmaps

Loop Source	Mean MAE ↓				Median MAE ↓			
	5%	15%	30%	50%	5%	15%	30%	50%
GT-Covisibility	0.088	0.226	0.361	0.408	0.099	0.259	0.369	0.385
None	0.090	0.239	0.387	0.441	0.100	0.279	0.415	0.427
SeqVLAD + UFM	<u>0.087</u>	<u>0.226</u>	<u>0.361</u>	<u>0.409</u>	<u>0.099</u>	<u>0.260</u>	<u>0.370</u>	<u>0.382</u>

((b)) Ground Truth Pixel Correspondences

Loop Source	Mean MAE ↓				Median MAE ↓			
	5%	15%	30%	50%	5%	15%	30%	50%
GT-Covisibility	0.073	0.174	0.262	0.299	0.081	0.174	0.246	0.269
None	0.078	0.192	0.296	0.337	0.088	0.200	0.285	0.309
SeqVLAD + UFM	<u>0.073</u>	<u>0.177</u>	<u>0.270</u>	<u>0.308</u>	<u>0.082</u>	<u>0.178</u>	<u>0.256</u>	<u>0.279</u>

Navigation Benchmarks

We compare PixelLoop's navigation performance against **Image-Relative (GNM, GNM-Adapted)** and **Object-Relative (ObjectReact)** topological baselines, both with and without loop closures.

In **GNM-Adapted**, we constrain the localised image selection to the dijkstra path from the goal image to the start image (weighted by GNM-predicted distances)

We also compare navigation performance with and without pixel-level loop closures. (**MASt3R-Nav**)

Navigation Benchmarks

Bold: Highest Overall

Underlined: Highest non-GT score

Method	Loop Source	SR $\uparrow$	SPL-S $\uparrow$	SPL-A $\uparrow$	SSPL $\uparrow$
PixelLoop (Ours)	GT-Covisibility	54.79	91.70	50.25	58.39
	None	35.62	80.45	28.65	40.70
	SeqVLAD + UFM	<u>68.49</u>	<u>91.15</u>	<u>62.43</u>	<u>73.97</u>
GNM (HM3D) [2]	GT-Covisibility	24.66	70.50	17.38	18.40
	None	20.55	64.80	13.32	15.95
	SeqVLAD + UFM	27.40	76.97	21.09	22.44
GNM-Adapted (HM3D)	GT-Covisibility	32.88	83.56	27.47	28.01
	None	17.81	75.28	13.41	16.95
	SeqVLAD + UFM	32.88	83.97	27.61	28.56
ObjectReact [1]	GT-Covisibility	63.01	65.21	41.09	65.91
	None	58.90	63.21	37.23	38.77
	SeqVLAD + UFM	67.12	69.37	46.82	68.35

Real-World Demonstration

- We deploy PixelLoop on a P-3DX mobile robot equipped with a RGB-D Intel RealSense camera in indoor environments.
- We communicate with the robot by publishing and subscribing to ROS topics, while our pipeline runs on a HPC cluster.
- **Computational Performance:**
 - **Real-time navigation: 4.8 Hz** online inference on a single **NVIDIA RTX A4000** GPU.
 - **Memory footprint: 4.6 GB VRAM** during navigation.
 - **MASt3R inference: 173 ms** for depth estimation and dense correspondence prediction.
 - **Costmap generation: 6.7 ms** for online query costmap filling at 320×240 resolution.
 - **Offline preprocessing:** SeqVLAD retrieval and UFM loop verification performed offline.
 - **Map preprocessing:** For a **350-frame**, **~100 m** mapping trajectory, goal-conditioned preprocessing takes **~1 minute per goal**, including **~25 s** for loop-aware shortest-path computation.

Real-World Demonstration

Sequential mapping run (4x)

Top-Down View

Pixel-level Loop Closure

RGB Feed



Summary

- Introduced **PixelLoop**, a pixel-space visual navigation framework with **dense pixel-level loop closures**.
- Unified disjoint trajectories and independent mapping runs **without requiring global metric reconstruction**.
- Achieved higher success rates and better path efficiency than image-level and object-level baselines in simulation.
- Validated the approach through real-world robot deployments.

Limitations

- **Performance depends on reliable pixel correspondences;** reflective, repetitive, or textureless scenes can reduce geometric accuracy.
- **Current implementation requires a discrete GPU** for real-time operation; further optimization is needed for edge devices.
- **Assumes largely static environments;** significant scene changes between mapping and navigation can degrade performance.

Future Work

- **Hierarchical semantic–pixel navigation:** Use objects, rooms, and landmarks for high-level planning while retaining pixel-level geometry for precise local navigation.
- **Multi-robot collaborative mapping:** Extend PixelLoop to support distributed exploration and merging of pixel-level topological maps.
- **Lifelong map adaptation:** Enable graph pruning, updating, and adaptation to long-term environmental changes.

Key References

01

ObjectReact

Garg et al. 2025, “Objectreact: Learning object-relative control for visual navigation,” in Conference on Robot Learning. PMLR

02

GNM

D. Shah et al., “Gnm: A general navigation model to drive any robot,” in IEEE International Conference on Robotics and Automation (ICRA), 2023.

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RoboHop

Garg et al., “Robohop: “ Segment-based topological map representation for open-world visual navigation,” in 2024 IEEE International Conference on Robotics and Automation (ICRA)

04

NoMaD

Sridhar et al., “Nomad: Goal masked diffusion policies for navigation and exploration,” in 2024 IEEE International Conference on Robotics and Automation (ICRA)

05

MASt3R-Nav

Garg et al., “Mast3r-nav: Waypixel navigation in relative 3d maps,” in International Conference on Robotics and Automation (ICRA), 2026.

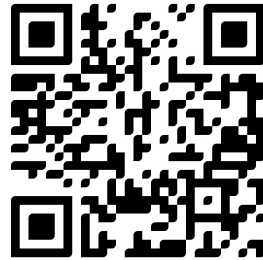
06

SeqVLAD

R. Mereu et al., “Learning sequence representations for visual place recognition,” in IEEE International Conference on Robotics and Automation (ICRA), 2022.

Thank You

Any Questions?



Project Page